BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: MRD-10TH 080523 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section – A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** MRD-10TH 080523 (A)
VERSION: B

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e A, B, C or D.

Q1. If the length of the pendulum become doubled then its time period becomes ..... times.

(A) $1/\sqrt{2}$
(B) $\sqrt{2}$
(C) $\sqrt[3]{2}$
(D) None

Q2. Human ear cannot hear the sound having frequency greater than .....

(A) 20Hz
(B) 2000Hz
(C) 20,000 Hz
(D) None

Q3. The human eye forms the image of an object at its .....

(A) Iris
(B) Pupil
(C) Cornea
(D) Retina

Q4. The phenomenon of total internal reflection is used in transmission of signal through.

(A) Electric wire
(B) Optical fiber
(C) Radio
(D) None

Q5. Origin of energy from the sun and stars is.

(A) Fission
(B) Fusion
(C) Radioactivity
(D) None

Q6. An echo occurs when sound wave is .....

(A) Absorbed
(B) Reflected
(C) Refracted
(D) Transmitted

Q7. The magnification of a refracting telescope is MQ = ...

(A) fo + fe
(B) fo - fe
(C) −fo/fe
(D) −fe/fo

Q8. The equivalent capacitance of a capacitors is always greater than their individual capacitance when they are connected in.

(A) Series
(B) Parallel
(C) Separate
(D) None

Q9. One kilowatt hour (1kwh) = ...

(A) 3.6×10^6J
(B) 1000J
(C) 3.6×10^3J
(D) 0.36hr

Q10. A convex lens with focal length 8 cm has the power of lens.

(A) 2.05D
(B) 4D
(C) 12.5D
(D) 16D

Q11. A step up transformer increases .....

(A) Power
(B) Energy
(C) Current
(D) Voltage

Q12. The Boolean equation for "AND" gate is.

(A) A.B
(B) A+B
(C) A+B with overline
(D) A.B with overline

Section – B

Time Allowed: 2.40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks:-

(i) Define longitudinal waves. Give three examples.

(ii) Why sound produced by simple pendulum is not heard?

(iii) What is speed of light in water having index of refraction as 1.33?

(iv) Are rearview mirror used in cars concave or convex?

(v) Normally objects with large number of electrons are electrically neutral why?

(vi) What is electrical resistance? Write any three factors on which it depends?

(vii) Define mutual induction and write its unit.

(viii) Differentiate between analogue and digital electronics.

(ix) A sound wave of wave length 1.7 x 10^{-2}m is travelling with velocity of 343.4 m/s calculate the frequency of sound.

(x) How electrostatic painting is better than conventional spray painting?

(xi) Draw truth table and symbol of "NAND" and "OR" gates.

Section – C

Maximum Marks: 21

Note: Answer any three (3) questions. All questions carry equal marks:-

Q3.

(a) Show that mass spring system execute simple harmonic motion.

(b) A mass hung from a spring vibrates 15 times in 12S. Calculate the frequency and time period of vibration.

Q4.

(a) Derive spherical mirror formula (1/f = 1/p + 1/q).

(b) A 1.50cm high object is placed 20cm from a concave mirror with radius of curvature 30cm find the position of image formed

Q5.

(a) Explain the force on a current carrying coil in a magnetic field.

(b) A 1.5cm long wire carries a current of 5A at right angle to a uniform magnetic field of magnitude 0.04T. Calculate the force exerted on the wire.

Q6.

(a) Describe the construction and working of electron gun.

(b) What are the algebraic Boolean expressions to represent the output of "AND" "OR" "NAND" and "NOR" gates.